

SCOPE DRIFT DETECTION IN DEFENSE ACQUISITION

Connecting GAO Program Failure Evidence to Operational Data Capture and System Info-Data Benchmarking

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Abstract

This concept note does two things. First, it answers a specific empirical question raised in peer review: Does the 22-year body of GAO annual weapon systems assessments support characterizing MDAP failure as primarily rooted in premature system definition or in execution failure? The evidence is examined directly. Second, it uses that finding to motivate and specify a Scope Drift Detection (SDD) architecture — a four-layer operational data capture and benchmarking framework that monitors system processes and AI behaviors in real time, surfaces emerging scope drift before it propagates into the info-data baseline, and provides program managers with a continuous, quantified signal that the operational envelope is evolving beyond what was formally defined.

The GAO evidence validates John's position: the dominant failure mode is execution — specifically, failure to effectively complete system information definition and realization at sufficient fidelity and maturity. The SDD architecture proposed here does not dispute that finding. It extends it: if execution failure is the problem, operational telemetry fed into a live info-data benchmark is the earliest-warning detection system available.

1. What GAO Actually Says About Failure Modes

1.1 Twenty-three years of consistent findings

GAO has published annual weapon systems assessments since 2002. The findings are remarkably consistent across administrations, acquisition reform cycles, and policy overhauls. The 2025 assessment (GAO-25-107569), the 23rd in the series, opens with the same essential observation as its predecessors: DOD continues to struggle with delivering timely and effective solutions to the warfighter, and the reforms intended to change this have not.

The question worth examining carefully is what GAO actually attributes these failures to — because the answer has direct implications for where an intervention architecture should be positioned.

1.2 The GAO failure taxonomy

Synthesizing across the 2022–2025 assessments, the failure drivers documented by GAO cluster into five categories. The table below maps each to its system info-data dimension — the specific gap in requirements specifications, architecture descriptions, design descriptions, or interface documentation that the failure represents or produces.

Failure Category	GAO Evidence (2022–2025)	System Info-Data Dimension
Technology immaturity at commitment	Programs entered MTA / MDAP milestones with TRL levels below the threshold. None of the seven reviewed former MTA programs with immature technologies were ready for production when efforts ended (GAO-25-107569).	Requirements specifications and architecture descriptions written before technology is proven create false precision in the info-data baseline — the data says 'defined,' but the knowledge doesn't exist yet.
Knowledge gaps at key decision points	GAO found a 20-year-consistent pattern: the majority of MDAPs do not fully achieve the knowledge needed to inform key investment decisions before proceeding (GAO-22-105230).	Absence of verified technical performance data means architecture and design descriptions are speculative, not authoritative. The info-data structure exists, but its content has not been validated against physical reality.
Software and design execution failures	Supplier, software, and design problems drove schedule delays across the 2023 portfolio; the B-52 Radar Modernization Program added 34 months due to software and testing issues (GAO-23-106059; GAO-25-107569).	This is an execution failure in the strictest sense — the processes that should produce verified drawings, interface control documents, and software design descriptions broke down under complexity.
Linear development in speed pathways	MTA programs — designed for 5-year completion — are averaging 10+ years. Programs continue to slow, sequential development despite pathway intent (GAO-24-106831).	Linear sequencing of info-data definition activities prevents feedback between architectural definition and design-level findings. Problems discovered late require upstream info-data rework at maximum cost.
Leading practices not implemented	Few programs report plans for digital twinning, digital threads, or minimum viable product approaches — even future programs with the opportunity to do so from inception (GAO-25-107569).	Digital thread and digital twin adoption are precisely the infrastructure that makes system information data continuously queryable and testable. Its absence means the info-data baseline is static documentation rather than live governance.

Table 1. GAO-documented MDAP failure categories mapped to system info-data dimensions. Sources: GAO-22-105230, GAO-23-106059, GAO-24-106831, GAO-25-107569.

1.3 The verdict on premature definition vs. execution failure

The GAO evidence supports John's position with specificity. The dominant failure mode is not that programs define their systems too early — it is that they proceed to commitment milestones without having adequately completed the knowledge work that makes those commitments sound. In GAO's language, programs do not fully achieve the knowledge needed to inform key investment decisions before proceeding. That is an execution failure in the knowledge-generation and info-data-definition processes, not a timing failure.

Where timing does appear as a factor — as in programs that commit to fixed solution concepts before technology is proven — the underlying problem is still execution: the processes that should validate technology readiness (testing, demonstration, modeling) were not completed to the fidelity required. The commitment happened not because the schedule forced premature definition but because the knowledge-generation work was not done.

There is one partial exception worth noting. GAO's 2025 finding that future programs do not plan to use leading practices from inception — before they become locked into rigid requirements, budgets, and development approaches — does gesture toward a timing problem. But even here, the issue is that the leading practices (iterative development, digital twinning, minimum viable product approaches) are absent from the plan, not merely that they are applied too late. The execution failure precedes the lock-in.

The implication for the proposal: the SBD framing should be refined from 'preventing premature commitment' to 'ensuring that the knowledge-generation and info-data-definition processes are complete before commitment milestones.' This is a more defensible and empirically grounded claim, and it aligns SBD directly with the MOVOI knowledge-threshold logic that is already the framework's core mechanism.

2. From GAO Evidence to Architecture: The Scope Drift Problem

2.1 Why execution failure persists despite policy reform

If the failure mode is execution — failure to complete system info-data definition at sufficient fidelity — the natural question is why 23 years of GAO findings and multiple rounds of acquisition reform have not resolved it. The answer visible in the data is structural: programs lack a continuous, automated signal that their info-data baseline is drifting from operational reality. They discover execution failures at milestone reviews, test events, or production — all of which are far downstream of the point where intervention would be cost-effective.

This is where operational data capture becomes architecturally consequential. The system being developed generates observable behavior throughout its development and early operational life. That behavior is a continuous stream of evidence about whether the requirements specifications, architecture descriptions, and design descriptions that constitute the formal baseline accurately represent what the system actually does. When they diverge, that divergence is scope drift — and it is detectable before it becomes a cost or schedule event.

2.2 Scope drift defined

Scope drift, as used here, is the progressive divergence between the formally defined operational envelope — as expressed in requirements specifications, architecture descriptions, design descriptions, and interface documentation — and the actual behavior of the system and its constituent processes, including AI-driven processes, as observed in operation.

Scope drift is distinct from requirements change (which is a controlled, documented update to the formal baseline) and from defect detection (which identifies a specific failure against a current requirement). Scope drift is the pre-change signal: the observable indication that the system's behavior is evolving in ways that the current info-data baseline does not capture, before that evolution has been formally recognized and processed as a change.

In systems with AI components, scope drift has an additional dimension. AI agents, particularly those using large language model inference chains, exhibit emergent behaviors — behaviors that arise from the composition of individually validated capabilities and that were not explicitly specified or anticipated in the architecture description. These emergent behaviors can be beneficial, neutral, or hazardous. The critical property they share is that they fall outside the formally defined operational envelope, making them invisible to traditional requirements tracing and TPM monitoring.

GAO's finding that weapon systems are more complex and software-driven than ever, combined with the consistent failure to implement digital thread and digital twin practices, means most programs have no continuous signal of this divergence. They are flying blind between milestone reviews.

3. Scope Drift Detection Architecture

3.1 Design principles

The Scope Drift Detection (SDD) architecture is designed around four principles derived directly from the GAO failure taxonomy and the system info-data framework.

- Continuous over episodic: Detection must operate continuously, not at milestone intervals. The cost of discovering scope drift at a CDR is an order of magnitude higher than discovering it in the preceding operational data stream.
- Layered signal integration: No single signal is sufficient. Operational telemetry, info-data currency, AI behavioral entropy, and knowledge gap quantification must be integrated to distinguish genuine scope drift from noise.
- Upstream causation: The architecture monitors the processes that produce info-data items, not just the items themselves. A degrading AI MTBH trend is a leading indicator of scope drift before any requirements specification has been updated.

- Quantified thresholds: Every layer produces a measurable signal with defined drift thresholds, enabling program managers to brief scope stability as a quantified metric rather than a qualitative assessment.

3.2 The four-layer architecture

The SDD architecture operates across four interdependent layers. Each layer captures a distinct class of operational data, benchmarks it against a defined baseline, and produces a scope drift indicator. The layers are designed to cascade: a signal at L1 (operational telemetry) triggers re-evaluation at L2 (info-data currency), which may surface a signal at L3 (AI behavioral entropy), which triggers MOVOI re-evaluation at L4 (knowledge gap surface).

Layer	Data Capture Mechanism	Benchmark Signal	Scope Drift Indicator
L1 — Operational telemetry	System process execution logs; AI agent invocation traces; sensor output streams	Baseline operational envelope defined in requirements spec at PDR	Deviation in execution path frequency, duration, or parameter range from baselined envelope — flags emergent process behavior before it propagates to info-data artifacts
L2 — Info-data currency	RDSG continuous query against SysML v2 repository; SHACL shape validation on all info-data item updates	Approved technical baseline (architecture and design descriptions at CDR)	Semantic distance between current repo state and approved baseline; SHACL violation rate as a leading indicator of definition drift rather than a lagging compliance check
L3 — AI behavioral entropy	MTBH rolling window; Brier Score trend across inference cycles; Circuit Breaker trip log	MTBH threshold and Brier calibration bound established during TMRR	Degrading MTBH trend signals AI behavioral drift — the system is encountering scenarios outside its validated operational envelope, often the first observable sign that requirements scope has shifted
L4 — Knowledge gap surface	MOVOI re-evaluation triggered by L1/L2/L3 signals; uncertainty quantification on open design decisions	MOVOI value-of-information thresholds set at SBD decision gates	Rising MOVOI values on previously closed design decisions indicate that new operational data has reopened epistemic uncertainty — a precise, quantified indicator that the scope has evolved beyond what was baselined

Table 2. Scope Drift Detection architecture layers, capture mechanisms, benchmark signals, and drift indicators.

3.3 Layer 1 — Operational telemetry

L1 captures system process execution data: the actual sequences, durations, parameter ranges, and branching patterns observed in operation. For AI-assisted processes, this includes agent invocation logs, inference chain traces, and output confidence distributions. The baseline is the operational envelope defined in requirements specifications and architecture descriptions at PDR — the formal description of what the system was designed to do and how.

The L1 drift indicator is parametric: when observed process behavior deviates from the baselined envelope in frequency, duration, parameter range, or branching pattern, the deviation is flagged. Small deviations within defined tolerance bands are expected and not indicative of drift. Sustained or directional deviations — where the system consistently operates near or beyond the edge of its defined envelope — are the signal.

L1 is the most granular layer and the highest-volume data source. Its role in the architecture is early warning: it surfaces the first observable sign that the system's operational reality is diverging from its formal definition, before that divergence has produced a change request, a test failure, or a cost event.

3.4 Layer 2 — Info-data currency

L2 monitors the system info-data baseline itself — the living body of requirements specifications, architecture descriptions, design descriptions, interface control documents, and associated artifacts maintained in the SysML v2 repository. The RDSG's SHACL validation layer provides the continuous query mechanism: every update to the repository is validated against the shape constraints that define structural and semantic integrity for each info-data item class.

The critical innovation at L2 is treating the SHACL violation rate as a leading indicator of definition drift rather than a lagging compliance check. Traditional use of constraint validation is reactive: a violation is raised, reviewed, and resolved. SDD uses the trend in violation rates — before violations are resolved — as a signal that the info-data definition processes are under stress. A rising violation rate on architecture description shapes, for example, indicates that engineers are attempting to update the architectural baseline in ways that the current shape constraints do not accommodate. That is a direct observable of scope drift propagating into the info-data layer.

L2 also monitors semantic distance — the degree to which the current repository state diverges from the approved technical baseline. This is not a binary approved/unapproved check but a continuous measure, enabling programs to track the rate of drift rather than simply detecting it at a threshold crossing.

3.5 Layer 3 — AI behavioral entropy

L3 is specific to systems with AI components and directly extends the MTBH and Brier Score frameworks developed in the proposal. The insight is that AI behavioral degradation — measured as a declining MTBH trend or a worsening Brier calibration curve — is a leading indicator of scope drift in AI-integrated systems, not merely a reliability metric.

When an AI agent's MTBH begins to decline — when it encounters hallucinations more frequently within a rolling operational window — the most common cause is that the system is being presented with inputs or scenarios outside the distribution it was validated against. That is, the operational

envelope has evolved beyond the formal boundary defined in the architecture description. The AI's degrading performance is the first observable sign that the scope has shifted, often weeks or months before any formal change request reflects it.

The Circuit Breaker mechanism provides the enforcement layer: when the MTBH threshold is violated, AI-generated outputs are quarantined from the authoritative SysML v2 baseline until reviewed. This prevents unvalidated emergent behavior from silently corrupting the info-data repository — which is precisely the mechanism by which AI-driven scope drift becomes invisible until it surfaces as a test failure or fielding incident.

The Brier Score adds a second dimension: it measures not just whether the AI is correct but whether it accurately represents its own uncertainty. An AI component whose Brier calibration is degrading is becoming overconfident in domains where its reliability is declining — a particularly hazardous form of behavioral drift for safety-critical systems, since the system's self-reported confidence no longer accurately indicates when human review is warranted.

3.6 Layer 4 — Knowledge gap surface

L4 integrates the signals from L1, L2, and L3 to re-evaluate the program's knowledge state using the MOVIOI framework. The core logic is: if operational telemetry, info-data currency monitoring, or AI behavioral entropy signals indicate that the system is operating in ways not captured by the current baseline, then the program's formal knowledge state — the epistemic foundation for previously closed design decisions — may have been invalidated by new evidence.

MOVIOI re-evaluation at L4 is triggered by sustained signals from lower layers. It quantifies whether the value of acquiring additional information on previously closed decisions has risen above the decision threshold — that is, whether new operational data has reopened questions that the program believed settled. This is the most consequential drift indicator: not merely that the system is behaving differently from its specification, but that the specification itself rests on assumptions that operational data is actively disconfirming.

L4 closes the loop back to SBD. The set-based design discipline is not a one-time application at TMRR; it is a continuous epistemological monitor. When L4 signals that previously closed design decisions have been reopened by new operational evidence, SBD provides the framework for managing the resulting uncertainty without triggering a full program reset — exactly the scenario for which the deliberate-decision-delay discipline was designed.

4. Integration with RDSG and the Existing Architecture

4.1 SDD as RDSG extension, not replacement

The SDD architecture is not a parallel system. It is an operational data integration layer that extends the RDSG's governance infrastructure into the system's behavioral life. The RDSG governs the info-data baseline — what has been formally defined and approved. SDD monitors the gap between that baseline and observable system behavior, surfacing the signals that should trigger RDSG updates before program managers discover them through milestone reviews.

Architecturally, SDD feeds into the RDSG's closed-loop requirements evolution mechanism. The AI behavioral failures that the original proposal identifies as information signals for requirements refinement are, in the SDD framework, specifically the L3 MTBH and Brier Score signals at the point where they indicate the operational envelope has evolved. SDD makes explicit the data capture and benchmarking layer that produces those signals systematically, rather than relying on ad hoc incident reporting.

4.2 STPA integration point

The STPA analysis scope introduced in RDSG v2.0 provides the formal loss scenario framework that SDD L1 should be benchmarked against. STPA's Unsafe Control Actions (UCAs) define the boundary conditions under which system processes produce hazardous outcomes. The L1 operational envelope baseline should be structured around the control action parameter ranges that STPA analysis has determined to be safe — meaning that any L1 drift signal crossing a UCA boundary is immediately escalated, bypassing the standard drift tolerance protocol.

This integration resolves the gap identified in the alignment analysis: STPA and RDSG are not merely parallel activities that should be connected in the SEP. SDD provides the runtime integration mechanism — STPA defines the hazard-sensitive boundaries in the L1 baseline; RDSG continuously validates the info-data expressions of those boundaries; and SDD monitors whether operational behavior remains within them.

4.3 The GAO leading practices connection

GAO's finding that few programs plan to implement digital twinning and digital thread practices — despite their demonstrated value for speed and quality — now has a specific architectural answer. The SDD architecture is the operational instantiation of the digital thread concept as applied to scope integrity: a continuous, machine-readable connection between the formal info-data baseline and the system's observed behavior, threaded through the RDSG semantic infrastructure.

Programs that implement SDD will be able to demonstrate, at any milestone review, not just that their info-data baseline is current but that the gap between their baseline and operational reality has been continuously monitored, quantified, and managed. That is the evidentiary standard that GAO's leading practices assessment is effectively asking for — and that no program in the current portfolio is delivering.

5. Implications for the Proposal and Next Steps

5.1 What changes in the proposal's framing

- The SBD framing on Nunn-McCurdy is strengthened by grounding it in GAO's knowledge-based acquisition evidence: SBD is not about avoiding premature commitment; it is about ensuring the knowledge-generation processes are complete, and SDD provides the continuous evidence that they are.
- The RDSG's scope gains an operational dimension. It is not only a governance framework for the pre-fielding info-data baseline; it is the anchor point for a live operational monitoring system that connects fielded system behavior back to the formal definition infrastructure.
- MTBH and the Brier Score gain a new application context: as scope drift leading indicators in AI-integrated systems, not only as standalone AI reliability metrics. This substantially elevates their strategic importance.
- The STPA integration gap identified in the alignment analysis has a specific resolution path through SDD's L1 architecture.

5.2 Prototype implications

The three-phase implementation roadmap maps directly onto SDD layer readiness. Phase 1 (Digital Authoritative Source of Truth via SysML v2/KerML) establishes the info-data baseline that L2 benchmarks against. Phase 2 (Virtual Integrated System via co-simulation) is the environment in which L1 operational telemetry is first generated, and L3 AI behavioral signals are first calibrated. Phase 3 (Physical HiL) transitions all four layers to real-world operational data, completing the loop from formal definition to observed behavior.

A working prototype that demonstrates all four SDD layers operating in real time — with quantified drift indicators updating against a live SysML v2 baseline — would be a genuinely differentiated capability. No current MDAP program office has this. It directly answers the leading practice gaps GAO has documented for 23 consecutive years.

5.3 The open research question

John's instinct that an AI synthesis of the full GAO report corpus would likely confirm execution failure as the dominant mode — rather than premature system definition — is well-founded based on the 2022–2025 evidence reviewed here. The more interesting research question is narrower: for the subset of programs that experienced Nunn-McCurdy cost breaches specifically, what does the GAO record show about the point in the development lifecycle at which the causal failure occurred, and was there any detectable operational signal in the period preceding the breach?

That question is answerable from the existing GAO corpus with targeted analysis. It would either confirm that continuous operational monitoring of the type SDD provides would have surfaced early signals or identify that the failure modes were at stages (e.g., early technology investment decisions) where SDD would not have been operational yet. Either finding sharpens the proposal's positioning.

Recommended action: Run a targeted analysis of the Nunn-McCurdy breach programs documented in GAO Table 1 historical data — specifically the Columbia-class, Sentinel, and B-52 RMP cases — to identify whether the documented failure causes had observable precursors of the type SDD L1 or L3 that would capture. This is a 2–3 day research effort and would produce a definitive answer to John's empirical question.